

1. (a) On C_1 : $z = r \exp(i\alpha)$, $dz = dr \exp(i\alpha)$ and

$$\begin{aligned} I_1 = \int_{C_1} f(z) dz &= \int_0^R r^n \exp(in\alpha) dr \exp(i\alpha) \\ &= \frac{R^{n+1}}{n+1} \exp(i(n+1)\alpha) \end{aligned}$$

On C_2 : $z = R \exp(it)$, $dz = iR \exp(it)dt$ and

$$\begin{aligned} I_2 = \int_{C_2} f(z) dz &= \int_\alpha^\beta R^{n+1} \exp(i(n+1)t) i dt \\ &= \frac{R^{n+1}}{n+1} [\exp(i(n+1)\beta) - \exp(i(n+1)\alpha)] \end{aligned}$$

On C_3 : $z = r \exp(i\beta)$, $dz = dr \exp(i\beta)$ and

$$\begin{aligned} I_3 = \int_{C_3} f(z) dz &= \int_R^0 r^n \exp(in\beta) dr \exp(i\beta) \\ &= -\frac{R^{n+1}}{n+1} \exp(i(n+1)\beta) \end{aligned}$$

As expected from Cauchy's theorem

$$I = \oint_C f(z) dz = I_1 + I_2 + I_3 = 0$$

- (b) To show that $f(z) = z^{*n}$ is not analytic, we can e.g. argue by contradiction. If $f(z)$ is analytic, then so is

$$\begin{aligned} g(z) &= z^n f(z) \\ &= z^n z^{*n} = |z|^{2n} \end{aligned}$$

The latter is constant along the circle $|z| = \text{const}$ and non-constant in the radial direction, at any $z \neq 0$. Hence for $z \neq 0$ both $g(z)$ and $f(z)$ are not differentiable. Therefore $f(z)$ is nowhere analytic.

On C_1 : $z = r \exp(i\alpha)$, $dz = dr \exp(i\alpha)$, $z^* = r \exp(-i\alpha)$ and

$$\begin{aligned} I_1 = \int_{C_1} f(z) dz &= \int_0^R r^n \exp(-in\alpha) dr \exp(i\alpha) \\ &= \frac{R^{n+1}}{n+1} \exp(i(1-n)\alpha) \end{aligned}$$

On C_2 : $z = R \exp(it)$, $dz = iR \exp(it)dt$, $z^* = R \exp(-it)$ and

$$I_2 = \int_{C_2} f(z) dz = \int_\alpha^\beta R^{n+1} \exp(i(1-n)t) i dt$$

When $n = 1$:

$$I_2 = R^2 i (\beta - \alpha)$$

When $n > 1$:

$$I_2 = \frac{R^{n+1}}{1-n} [\exp(i(1-n)\beta) - \exp(i(1-n)\alpha)]$$

On C_3 : $z = r \exp(i\beta)$, $dz = dr \exp(i\beta)$, $z^* = r \exp(-i\beta)$ and

$$\begin{aligned} I_3 = \int_{C_3} f(z) dz &= \int_R^0 r^n \exp(-in\beta) dr \exp(i\beta) \\ &= -\frac{R^{n+1}}{n+1} \exp(i(1-n)\beta) \end{aligned}$$

Overall, when $n = 1$:

$$I = \oint_C f(z) dz = I_1 + I_2 + I_3 = R^2 i (\beta - \alpha)$$

When $n > 1$:

$$I = R^{n+1} \left(\frac{1}{1-n} - \frac{1}{1+n} \right) [\exp(i(1-n)\beta) - \exp(i(1-n)\alpha)]$$

This is non-zero, except when $(1-n)(\beta - \alpha) = 2k\pi$: a non-analytic function may or may not integrate to zero around individual closed paths.

2. C is the circle of radius 3 centered at the origin.

(a)

$$I = \oint_C \exp(2z) dz$$

The integrand is analytic everywhere so $I = 0$ by Cauchy's theorem.

(b)

$$I = \oint_C \frac{\exp(2z)}{z+4} dz$$

The integrand has a singularity at $z = -4$ which is outside C . On and within C the integrand is everywhere analytic so $I = 0$ by Cauchy's theorem.

(c)

$$I = \oint_C \frac{\exp(2z)}{(z+i)^3} dz$$

We can apply the Cauchy integral formula for derivatives

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z-z_0)^{n+1}} dz$$

with $f(z) = \exp(2z)$, $n = 2$, $z_0 = -i$ (which is inside C).

Using $f^{(2)}(z) = f''(z) = 4 \exp(2z)$:

$$I = \frac{2\pi i}{2!} 4 \exp(-2i) = 4\pi i (\cos 2 - i \sin 2)$$

3. C is the square of side 2, centre at the origin and sides parallel to the x - and y -axes.

(a)

$$I = \oint_C \frac{\sinh z}{z - i/2} dz$$

We can apply the Cauchy integral formula

$$\oint_C \frac{f(z)}{z - z_0} dz = 2\pi i f(z_0)$$

with $f(z) = \sinh z$ and $z_0 = i/2$ (which is inside C):

$$I = 2\pi i \sinh(i/2) = 2\pi i \sin(1/2) = -2\pi \sin(1/2)$$

(b)

$$I = \oint_C \frac{\cos(z+1)}{z^4} dz$$

We can apply the Cauchy integral formula for derivatives

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz$$

with $f(z) = \cos(z+1)$, $n = 3$, $z_0 = 0$ (which is inside C).

Using $f^{(3)}(z) = f'''(z) = \sin(z+1)$:

$$I = \frac{2\pi i}{3!} \sin 1 = \frac{\pi i}{3} \sin 1$$

(c)

$$I = \oint_C \frac{\sinh z}{(z^2 - 4)^2} dz$$

The integrand has singularities at $z^2 = 4 \Rightarrow z = \pm 2$, which is outside C . On and within C the integrand is everywhere analytic so $I = 0$ by Cauchy's theorem.

4. If $f(z)$ is analytic on and within a simple closed contour C , and z_0 is inside C , then we have the Cauchy integral formula for the n -th derivative

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz$$

If C encloses the origin and we choose $z_0 = 0$, then

$$f^{(n)}(0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{z^{n+1}} dz$$

If $f(z) = \exp(-iz)$, then $f^{(n)}(z) = (d/dz)^n \exp(-iz) = (-i)^n \exp(-iz)$. At $z = 0$ this becomes $f^{(n)}(0) = (-i)^n$. Hence

$$(-i)^n = \frac{n!}{2\pi i} \oint_C \frac{\exp(-iz)}{z^{n+1}} dz$$

Let now C be the circle $|z| = r$. For z on C , change variable to $z = r \exp(i\theta) = r \cos \theta + ir \sin \theta$, $dz = ir \exp(i\theta) d\theta$, $0 \leq \theta \leq 2\pi$. Substituting in:

$$\int_0^{2\pi} \exp(-i(r \cos \theta + ir \sin \theta)) \frac{ir \exp(i\theta)}{r^{n+1} \exp(i(n+1)\theta)} d\theta = \frac{2\pi i}{n!} (-i)^n$$

Rearranging:

$$\int_0^{2\pi} \exp(r \sin \theta) \exp(-i(n\theta + r \cos \theta)) d\theta = \frac{2\pi r^n}{n!} (-i)^n$$

Writing $(-i)^n = \exp(-in\pi/2)$ and taking real and imaginary parts of the two sides:

$$\begin{aligned} \int_0^{2\pi} \exp(r \sin \theta) \cos(n\theta + r \cos \theta) d\theta &= \frac{2\pi r^n}{n!} \cos(n\pi/2) \\ \int_0^{2\pi} \exp(r \sin \theta) \sin(n\theta + r \cos \theta) d\theta &= \frac{2\pi r^n}{n!} \sin(n\pi/2) \end{aligned}$$

5. (a) $f(z) = z \cos z$. Using the Taylor series for $\cos z$ about $z = 0$:

$$\begin{aligned} f(z) &= z \cos z \\ &= z \left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots \right) \\ &= z - \frac{z^3}{2!} + \frac{z^5}{4!} - \frac{z^7}{6!} + \dots \end{aligned}$$

- (b) $f(z) = \frac{\exp z}{1+z}$. We use the Taylor series for the exponential function about $z = 0$ and the binomial series for $(1+z)^{-1}$, which is absolutely convergent for $|z| < 1$:

$$\begin{aligned} f(z) &= \frac{\exp z}{1+z} \\ &= \left(1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \dots \right) (1 - z + z^2 - z^3 + z^4 - \dots) \\ &= 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} \\ &\quad - z - z^2 - \frac{z^3}{2!} - \frac{z^4}{3!} + z^2 + z^3 + \frac{z^4}{2!} \\ &\quad - z^3 - z^4 + z^4 + \dots \\ &= 1 + \frac{1}{2!} z^2 + \left(\frac{1}{3!} - \frac{1}{2!} \right) z^3 + \left(\frac{1}{4!} - \frac{1}{3!} + \frac{1}{2!} \right) z^4 + \dots \\ &= 1 + \frac{1}{2} z^2 - \frac{1}{3} z^3 + \frac{3}{8} z^4 + \dots \end{aligned}$$

6. Find the Laurent series below.

(a)

$$f(z) = \frac{1}{(z+1)(z-1)} \quad \text{for } 0 < |z-1| < 2$$

Let $w = z - 1$:

$$\begin{aligned} f(z) &= \frac{1}{(z+1)(z-1)} \\ &= \frac{1}{w(w+2)} \\ &= \frac{1}{2w(1 + \frac{w}{2})} \\ &= \frac{1}{2w} \left[1 - \frac{w}{2} + \left(\frac{w}{2}\right)^2 - \left(\frac{w}{2}\right)^3 + \dots \right] \\ &= \frac{1}{2w} - \frac{1}{4} + \frac{1}{8}w - \frac{1}{16}w^2 + \dots \end{aligned}$$

The binomial series is valid provided $|\frac{w}{2}| < 1$ or $|z-1| < 2$ as required.

(b)

$$f(z) = \frac{1}{(z+1)(z-1)} \quad \text{for } |z| > 1$$

Let $w = 1/z$:

$$\begin{aligned} f(z) &= \frac{1}{(z+1)(z-1)} \\ &= \frac{1}{\left(\frac{1}{w} + 1\right)\left(\frac{1}{w} - 1\right)} \\ &= \frac{w^2}{(1+w)(1-w)} \\ &= \frac{w^2}{1-w^2} \\ &= w^2 [1 + w^2 + w^4 + \dots] \\ &= w^2 + w^4 + w^6 + \dots \end{aligned}$$

The binomial series is valid provided $|w^2| < 1$ or $|z| > 1$ as required.

(c)

$$f(z) = \frac{1}{z(z-1)} \quad \text{for } |z| > 1$$

Let $w = 1/z$:

$$\begin{aligned} f(z) &= \frac{1}{z(z-1)} \\ &= \frac{1}{\frac{1}{w} \left(\frac{1}{w} - 1 \right)} \\ &= \frac{w^2}{1-w} \\ &= w^2 [1 + w + w^2 + \dots] \\ &= w^2 + w^3 + w^4 + \dots \end{aligned}$$

The binomial series is valid provided $|w| < 1$ or $|z| > 1$ as required.

(d)

$$f(z) = \frac{\sin z}{z^3} \quad \text{for } |z| > 0$$

Use the Taylor series for $\sin z$ about the origin:

$$\begin{aligned} f(z) &= \frac{\sin z}{z^3} \\ &= \frac{z - \frac{z^3}{3!} + \frac{z^5}{5!} - \dots}{z^3} \\ &= \frac{1}{z^2} - \frac{1}{3!} + \frac{z^2}{5!} - \dots \end{aligned}$$

This is valid for $|z| > 0$.
